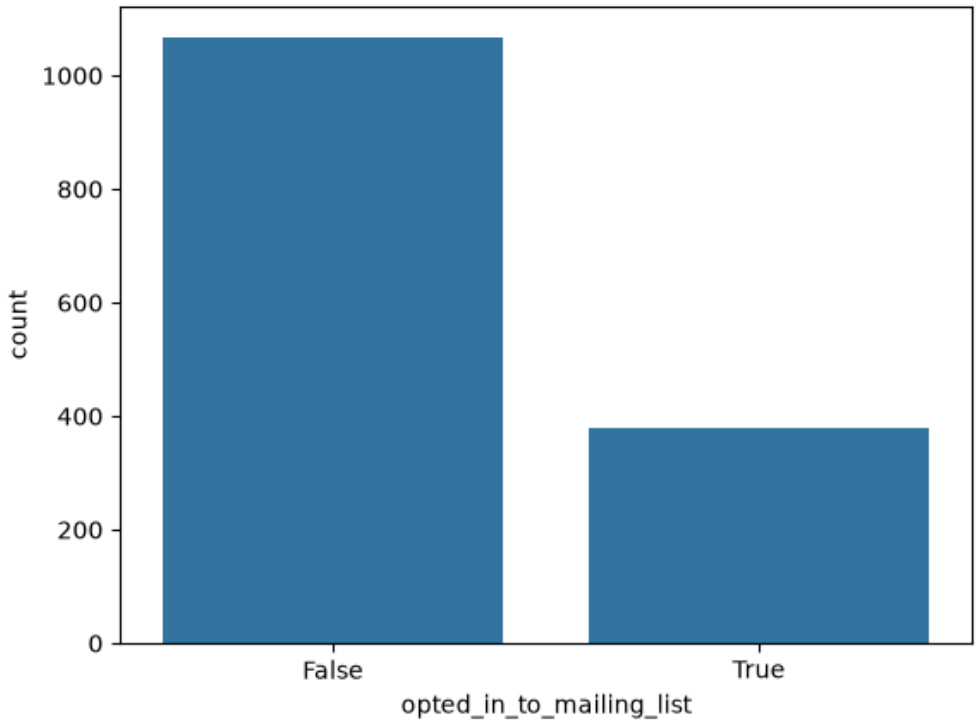
Relax Data Science Challenge

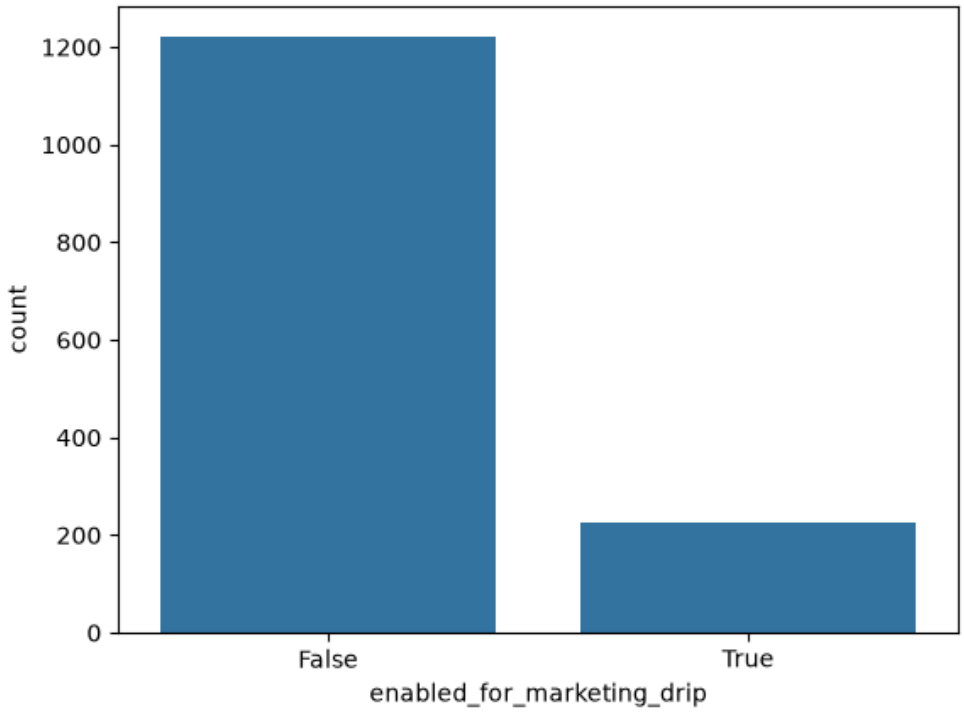
There can be many things that determine if a user will continue using a product. However, given the data I was provided, I believe that there are some key factors to help determine if a user will be more likely to adopt the product. As after filtering out the user login records to users who are logging at least three times a week, it appears that out of the 12,000 users that are signed up only 1445 users can be considered heavy users. Out of these heavy users, the most common thing among them is that they are a part of an organization. As they make up around half of these filtered users. Because of this, it seems to be pointless to send out marketing emails given that our most heavy users are business and organizations. However it also appears that there is not one clear organization who is using the product over another. The same can also be said about the trend regarding when people sign up for the product. As simply put, the longer a user has been using the product, the more likely they are to be a heavy user. As this is heavily demonstrated in recent months. Therefore to summarize, the most likely factor that determines if a user will adopt the product is if they join an organization who is currently using the product

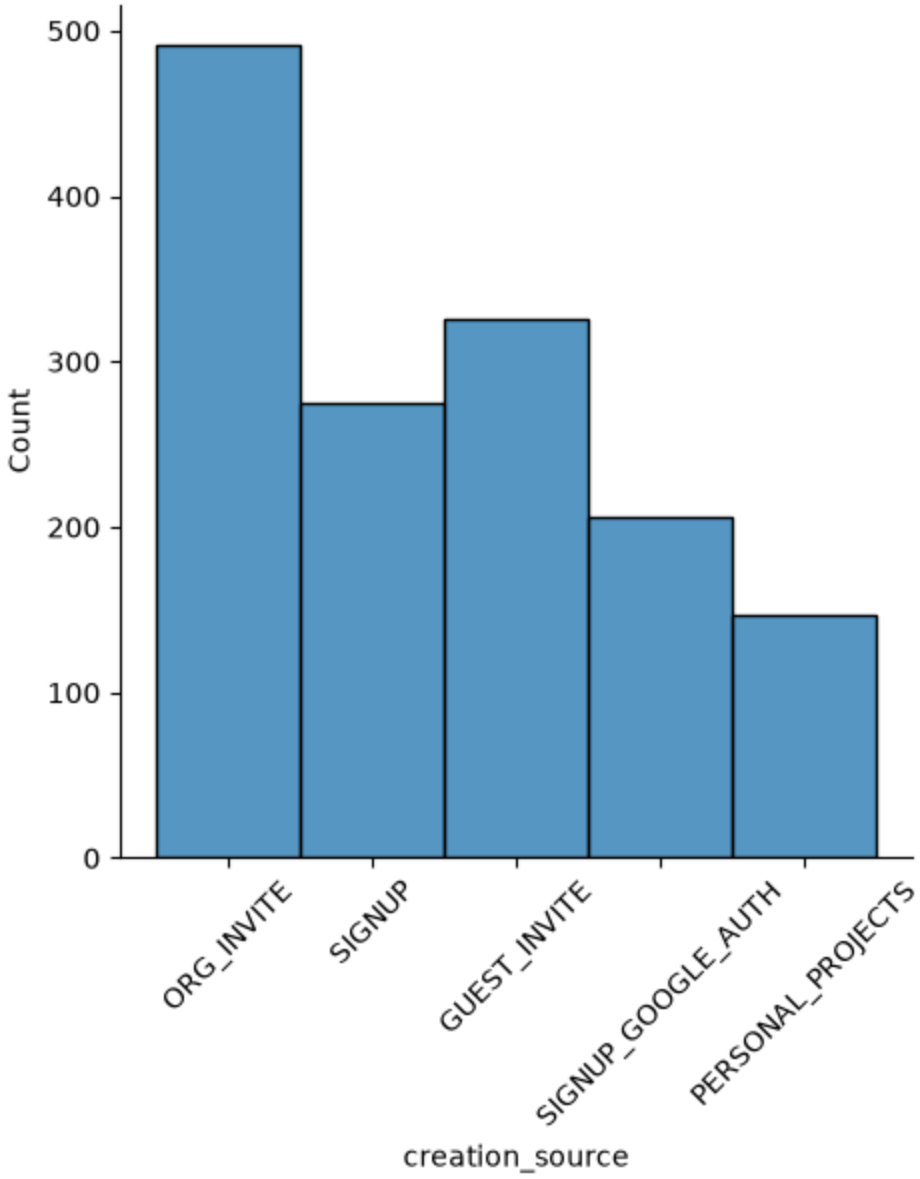
Some factors that I took into consideration was that that user had to use the product 3 times in any given week as this helps show that they are willing to use the product and not just trying it out temporarily. Some other investigation that I tried was trying to find a trend among org ids and referrer ids however given the large total number between both, this proved difficult to perform.

Further research that can be helpful is what industry the organization is in to help determine further patterns. Along with determining where these users are located to help determine if there are any geographical trends

Figures / Code Snippets






500
400
300
200
100
0
Count
ORG_INVITE
SIGNUP
GUEST_INVITE
SIGNUP_GOOGLE_AUTH
PERSONAL_PROJECTS
creation_source

```
result['invited_by_user_id'].value_counts().head(10)
```

[145] ✓ 0.0s Python

```
invited_by_user_id
2354.0    4
7358.0    3
9726.0    3
10163.0   3
9510.0    3
10628.0   3
2771.0    3
7882.0    3
4019.0    3
3623.0    3
Name: count, dtype: int64
```

```
heavy_user_list['org_id'].value_counts().head(10)
```

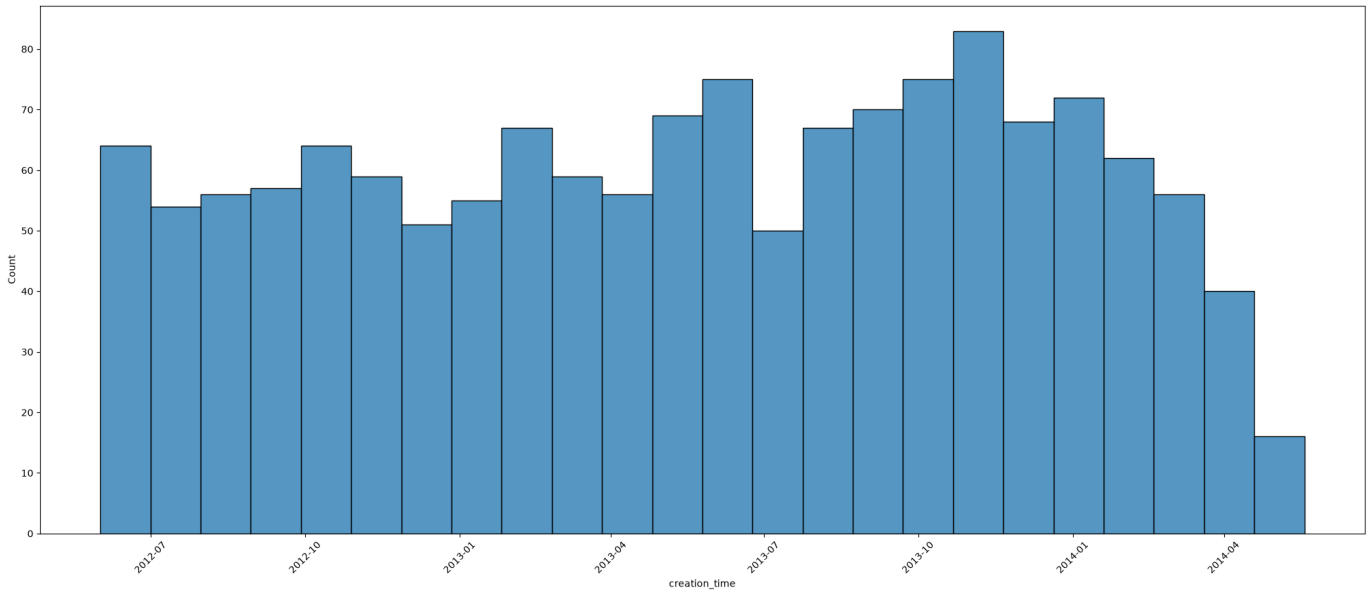
[144] ✓ 0.0s Python

```
org_id
7     16
1     14
2     13
9     13
13    13
3     13
62    12
5     11
4     11
0     11
Name: count, dtype: int64
```

```
heavy_user_list['creation_source'].value_counts()
```

[148] ✓ 0.0s Python

```
creation_source
ORG_INVITE           491
GUEST_INVITE         326
SIGNUP               275
SIGNUP_GOOGLE_AUTH   206
PERSONAL_PROJECTS    147
Name: count, dtype: int64
```


Count
0
10
20
30
40
50
60
70
80
2012-07
2012-10
2013-01
2013-04
2013-07
2013-10
2014-01
2014-04
creation_time